

Ebrahim Karam

Controls & Software Engineer

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PROFILE

Controls and Software Engineer with 5+ years in industrial automation, **PLC programming (Structured Text, TwinCAT, Codesys)**, and HMI development for complex electromechanical systems. Skilled in system integration, embedded C++, and real-time control.

WORK EXPERIENCE

- Animax Designs** Nashville, TN, USA
Jul 2022 – Jul 2024
Controls & Software Engineer
Architected and developed real-time control logic for complex animatronic figures using **Structured Text on Beckhoff (TwinCAT) PLCs**, implementing robust fault handling, alarms, and system interlocks to ensure safe and reliable operation.
Engineered low-latency motor control firmware in **Modern C++ for ESP32 microcontrollers**, integrating embedded systems with PLC logic for precise, real-time finger gesture control.
Designed and deployed intuitive **HMI/GUI interfaces in C# and Perspective Ignition**, enabling real-time system monitoring, control, alarm visualization, and remote diagnostics for attractions and animatronics.
Integrated animatronic systems with external Show Control Systems via industrial communication protocols, ensuring seamless system-level operation.
Configured local networking infrastructure and managed system bring-up, commissioning, and on-site validation for global installations, including high-profile projects in Sydney and Singapore.
- Third Coast Automation** Nashville, TN, USA
Jul 2024 – Oct 2024
Software Control Engineer (Contract)
Developed the core control application for a fuel polishing system using **Structured Text and Ladder Logic on WAGO PLCs (Codesys-based)**, managing sensors, actuators, and safety interlocks.
Designed an HMI for system monitoring and control using the native HMI tools within the Codesys environment.
Authored comprehensive verification and Factory Acceptance Test (FAT) documentation to support system commissioning and validation.
- 1220 Exhibits** Nashville, TN, USA
Oct 2024 – Jul 2025
Interactive Technician (Hourly)
Spearheaded the end-to-end deployment of embedded C systems and interfaced directly with production hardware (Arduino, ESP32, CONTROLLINO) for high-profile museum exhibits.
Participated in system bring-up, integration, and troubleshooting of complex electromechanical systems in live environments.
- Western Express** Nashville, TN, USA
Sep 2025 – Present
Full Stack Developer
Engineered high-throughput data parsing pipelines and automated API workflows, optimizing execution time and reliability using Python, Apache Airflow, and Javascript.
Architect and maintain scalable Ubuntu Linux environments, systemd services, and Nginx reverse proxies.
- AUB Aerosol Lab**, American University of Beirut Beirut, Lebanon
Oct 2017 – Dec 2020
Research Engineer
Designed and built custom embedded systems using **C and Arduino** to control experimental apparatus and collect data for research studies.
Developed a data retrieval system for an air pollution monitor, automating data collection and preliminary analysis in Python, running on a Lubuntu environment.

EDUCATION

- Carnegie Mellon University, Entertainment Technology Center (ETC)** Pittsburgh, PA
Master of Entertainment Technology
May 2022
- American University of Beirut (AUB)** Beirut, Lebanon
BE in Electrical and Computer Engineering
Sep 2013 – Sep 2017
Minor in Mathematics and Biomedical Engineering

SKILLS

Controls & Automation: IEC 61131-3 (Structured Text), PLC (Beckhoff TwinCAT, WAGO Codesys), HMI Development (C#, Perspective Ignition, Codesys), Industrial Communication Protocols (Modbus, EtherCAT, TCP/IP), System Architecture & Integration

Programming Languages: MODERN C++, C, PYTHON, C#, SQL, STRUCTURED TEXT, POWERSHELL

Embedded Systems & Tools: Microcontrollers (ESP32, Arduino), Linux (Ubuntu), Nginx, Git

Languages: English (Native/Bilingual), Arabic (Native/Bilingual), French (Limited Working)

AWARDS & SCHOLARSHIPS

- Gold Medal Winner, Serious Play Conference Jun 2022
Best Overall, Makeathon "Rethink the Rink" (Purple Team) Mar 2022
Recipient, Electronic Arts Diversity and Inclusion Award Academic Year 2021-2022
1st Prize, Beirut Startup Weekend by TechStars Oct 2017